17 EXAM PRACTICE

1 A precipitate of copper(II) hydroxide dissolves in aqueous ammonia. What is the formula of the complex ion formed?

A $[Cu(NH_3)_2]^{2+}$

B $[Cu(NH_3)_4]^{2+}$

C $[Cu(NH_3)_6]^{2+}$

D $[Cu(NH_3)_4(H_2O)_2]^{2+}$ [1]

(Total for Question 1 = 1 mark)

2 What is the electronic configuration of the iron cation that can form the complex ion $[Fe(CN)_6]^{4-}$? [The cyanide ligand has the formula CN^-.]

A [Ar] $3d^4\ 4s^2$

B [Ar] $3d^5\ 4s^0$

C [Ar] $3d^6\ 4s^0$

D [Ar] $3d^6\ 4s^2$ [1]

(Total for Question 2 = 1 mark)

3 The anti-cancer drug *cis*-platin has the general formula $Pt(NH_3)_2Cl_2$. In the human body, one of the chloride ions of *cis*-platin is replaced by one water molecule:

$$Pt(NH_3)_2Cl_2 + H_2O \rightarrow [Pt(NH_3)_2(H_2O)Cl]^+ + Cl^-$$

What are the oxidation numbers of platinum in these substances?

	Oxidation number in *cis*-platin	Oxidation number in the aqua complex
A	+2	+1
B	+2	+2
C	+4	+3
D	+4	+4

[1]

(Total for Question 3 = 1 mark)

4 A compound of chromium with the general formula $CrCl_3.6H_2O$ is dissolved in water. When an excess of aqueous silver nitrate is added, only one-third of the total chloride present is precipitated as AgCl.

What is the formula of the chromium complex present in $CrCl_3.6H_2O$?

A $[Cr(H_2O)_6]^{3+}$

B $[Cr(H_2O)_5Cl]^{2+}$

C $[Cr(H_2O)_4Cl_2]^{+}$

D $[Cr(H_2O)_3Cl_3]$ [1]

(Total for Question 4 = 1 mark)

5 Titanium has the electronic configuration of $1s^2\ 2s^2\ 3s^2\ 3p^6\ 3d^2\ 4s^2$.

Which of the following compounds is *unlikely* to exist?

A TiO

B TiO_2

C K_3TiF_6

D K_2TiO_4 [1]

(Total for Question 5 = 1 mark)

6 What are the colours of aqueous solutions containing $Cr_2O_7^{2-}$, CrO_4^{2-}, $[Cr(H_2O)_6]^{3+}$ and $[Cr(H_2O)_6]^{2+}$?

	$Cr_2O_7^{2-}$	CrO_4^{2-}	$[Cr(H_2O)_6]^{3+}$	$[Cr(H_2O)_6]^{2+}$
A	yellow	orange	green	blue
B	orange	yellow	green	blue
C	orange	yellow	blue	green
D	orange	green	yellow	blue

[1]

(Total for Question 6 = 1 mark)

7 Why is the hexaaquacopper(II) ion, $[Cu(H_2O)_6]^{2+}$, blue?

A The water ligands split the p-orbital energies and p–p electron transitions emit blue light.

B The water ligands split the d-orbital energies and d–d electron transitions absorb all but blue light.

C The water ligands split the p-orbital energies and p–p electron transitions absorb all but blue light.

D The water ligands split the d-orbital energies and d–d electron transitions emit blue light. [1]

(Total for Question 7 = 1 mark)

8 Scandium and titanium are both d-block elements, but only titanium is a transition metal.

(a) Complete the electronic configuration of each of these ions.

Sc^{3+} $1s^2 2s^2 2p^6$ .....

Ti^{3+} $1s^2 2s^2 2p^6$ ..... [2]

(b) Explain why titanium is a transition metal, but scandium is not. [2]

(c) What is the oxidation number of the d-block element in each of these compounds?

$Sc(OH)_3$ .....

$CaTiO_3$ ..... [2]

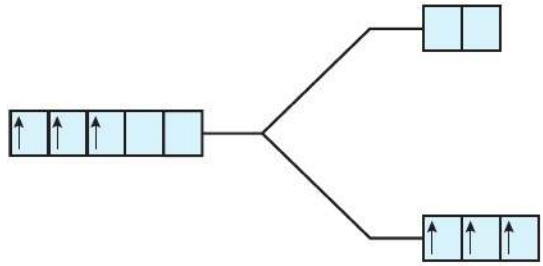
(Total for Question 8 = 6 marks)

9 (a) What is the meaning of the term 'ligand'? [2]

(b) (i) Give the formula of the complex with the name tetraaquadihydroxoiron(II). [1]

(ii) Give the name of the complex with the formula $[CoCl_4]^{2-}$. [1]

(c) The arrangement of d-electrons in the complex with the formula $[Cr(H_2O)_6]^{3+}$ can be represented using this diagram.



Explain, with reference to this diagram, why a solution containing this complex is coloured. [5]

(Total for Question 9 = 9 marks)

10 Most transition metal complexes have octahedral or tetrahedral shapes.

(a) Explain why the complex formed between one copper(II) ion and six water molecules has an octahedral shape. [2]

(b) Concentrated hydrochloric acid is added to a solution containing the complex in part (a). During the reaction that occurs, there is a colour change from blue to yellow.

(i) Explain why the number of ligands around the metal ion decreases during this reaction. [2]

(ii) State one other feature of the complex that changes during this reaction, and one feature that does *not* change. [2]

(c) A complex with a linear shape has the formula $[Ag(NH_3)_2]^+$, and is used in a test to distinguish between aldehydes and ketones.

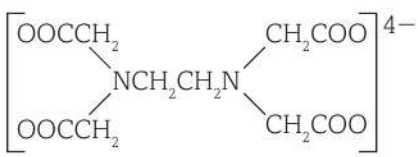
(i) Name this complex and the reagent that contains it. [2]

(ii) State the type of change that this complex undergoes when it reacts with an aldehyde, and the name of the metal species formed. [2]

(d) A complex with a different shape, used in cancer treatment, is *cis*-platin, and has the formula $[Pt(NH_3)_2Cl_2]$.

Draw the structure of *cis*-platin and state the name of its shape. [2]

(e) A multidentate ligand has a formula abbreviated to $EDTA^{4-}$. Its structure is:



(i) Explain why $EDTA^{4-}$ is described as a multidentate ligand. [1]

(ii) Draw the structure of $EDTA^{4-}$ showing the lone pairs of electrons used in the formation of bonds with a transition metal ion. [1]

(iii) Using the abbreviation EDTA, give the formula of the complex formed when $[Fe(H_2O)_6]^{3+}$ reacts with $EDTA^{4-}$. [1]

(Total for Question 10 = 15 marks)

11 The following sequence summarises some reactions, 1, 2, 3 and 4, of chromium compounds.

$$[Cr(H_2O)_6]^{3+} \xrightarrow{1} [Cr(H_2O)_3(OH)_3] \xrightarrow{2} [Cr(OH)_6]^{3-} \xrightarrow{3} CrO_4^{2-} \xrightarrow{4} Cr_2O_7^{2-}$$

(a) State the appearance of the reactant and product in reaction 1. [2]

(b) State the reagent used in both reactions 1 and 2. [1]

(c) With reference to oxidation numbers, identify which reaction is a redox reaction. [2]

(d) Describe the colour change in reaction 4. [1]

(e) Write an equation for reaction 4. [1]

(Total for Question 11 = 7 marks)

12 The heterogeneous catalyst used in the Contact process to manufacture sulfuric acid has the formula V_2O_5.

Explain, with the help of suitable equations and with reference to the oxidation states of vanadium, the steps in this catalysed reaction. [6]

(Total for Question 12 = 6 marks)